Question 6

In a hospital, there are various departments. Each doctor is associated with one department. A patient gets registered to a department under one doctor in that department. It may be required, that he/she may be treated by other doctors of same /different departments also. Doctors suggest various tests for the patients, those are noted along with their reports.

Marks: 2 **MSQ**

If we want to design a schema for the above system, what are the possible entities?

a) Report

b) Doctor

c) Patient

d) Test

Answer: b), c)
Explanation:
The system will be created for patients, so it is the most important entity.
A patient gets registered under a doctor, so Doctor is another important entity.
So, options (b) and (c) are the correct options.

Question 9

A tourism department operates boating facility at one of its ECO-PARK. There is a number of boats having different seating capacities. These boats are owned by a number of boatmen. A boatman may own more than one boat but a boat is owned by only one boatman. Each boat has an identity code. Tourist party may book boats according to thenumber of persons in a party. A party may book more than one boat or several parties may book same boat. Payment will be charged from the parties on the basis of number of persons, hours booked and boat types. Records of all rides are kept to charge boatmen a certain percentage of their income.
Which of the following(s) is (are) essential entities for designing the database?

Marks: 2 **MSQ**

a) `Boats`

b) `Department`

c) `Tourist party`

d) `Payment`

Answer: a), c)
Explanation: According to the problem description, there will be three entities: `Boats`, `Boatmen` and `Tourist Party`.
We do not need to store any information about the `Eco-park` and `Department` in our database.
`Payment` can be the relationship between `Boatmen` and `Tourist Party`. So, it is not an entity.
So, options (a) and (c) are correct.

Question 10

A tourism department operates boating facility at one of its ECO-PARK. There is a number of boats having different seating capacities. These boats are owned by a number of boatmen. A boatman may own more than one boat but a boat is owned by only one boatman. Each boat has an identity code. Tourist party may book boats according to thenumber of persons in a party. A party may book more than one boat or several parties may book same boat. Payment will be charged from the parties on the basis of number of persons, hours booked and boat types. Records of all rides are kept to charge boatmen a certain percentage of their income.

Identify the correct statement(s) about the database. *Marks: 2* **MSQ**

a) There will be a many-to-many relationship between `Tourist party` and `Boats`.

b) There will be a many-to-many relationship between `Boats` and `Boatmen`.

c) `Boat_Id` must be the foreign key of `Boatmen` table.

d) `Boatmen_ID` must be a foreign key in `Payment` relation.

Answer: a), d)

Explanation: According to the statement "A party may book more than one boat or several parties may book same boat." Hence, there will be a **many-to-many relationship** between `Tourist party` and `Boats`.

According to the statement "A boatman may own more than one boat but a boat is owned by only one boatman." So, there will be a **many-to-one relationship** between `Boats` and `Boatmen`.

`Boat_Id` must be the foreign key of the relationship table between `Boats` and `Boatmen`, not for the entity `Boatmen`.

`Payment` must be a **many-to-many relationship** table between `Tourist party` and `Boatmen`. So, `Boatmen_ID` must be a foreign key of `Payment` relation. Hence, options (a) and (d) are correct.

Question 5

In an Annual Sports, players individually can enroll their name or a team also can enroll for the sports. There can be many players in a team but one player can play only for one team. A player can participate in many games and a game can be played by multiple players. Each team has a unique id and name. Player also has a unique id and name. Each game has a unique name, type and result.

Which is (are) the correct statement(s) from the following? *Marks: 2* **MSQ**

a) Entity Team will not have any primary key.

b) CanPlay will be a one-to-many relationship between Team and Players.

c) G_name can be the foreign key of Participate relation between Players and Game.

d) Player_Id will be the foreign key of Game table.

Answer: b), c)

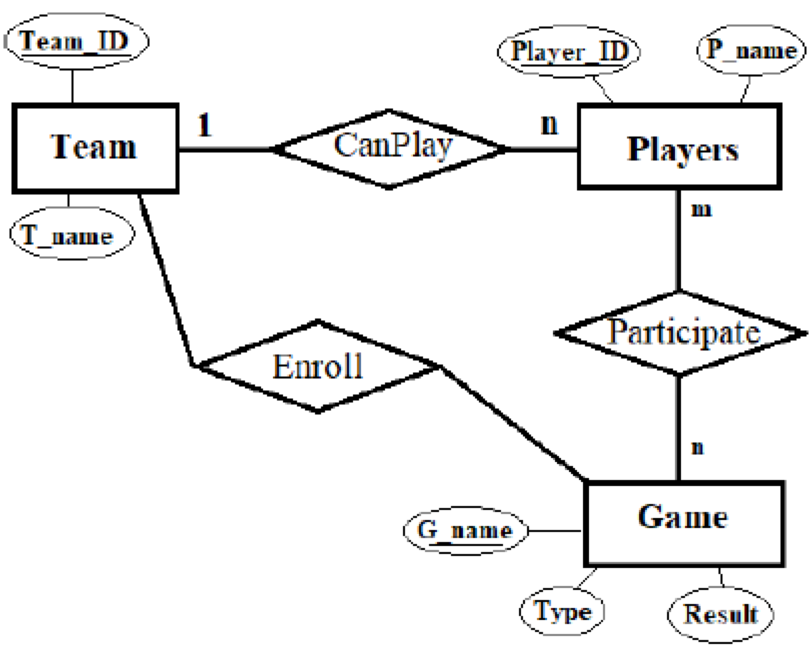
Explanation: Primary key of entity Team is Team_ID .

Player_Id can never be the foreign key of Game table as the relation between Game and Players are many-to-many.

So, options a) and d) are false statements.

There will be a one-to-many relationship between Team and Players.

G_name is the primary key of entity Game table. Hence, can be the foreign key of Participate relation between Players and Game.



Hence, option b) and c) are correct.

FD-Normalization

Question 1

Given below is the table **department_manager_employee**

department_manager_employee		
employee_id	**department**	**manager**
1101	Finance	Sam
1102	R&D	Rohit
1101	HR	Sam
1103	Retail	Rohit

Find out the functional dependencies that **do not** hold on the above table:

Marks: 2 **MSQ**

a) $employee\_id \rightarrow manager$

b) $employee\_id \rightarrow department$

c) $manager \rightarrow department$

d) $department \rightarrow manager$

Answer: b), c)

Explanation: A particular `manager` manages more than one department, hence $manager \rightarrow department$ is invalid on the above table.
A particular `employee_id` determines more than one department, hence $employee\_id \rightarrow department$ is also invalid on the above table.

Question 1

Consider the relation R(A, B, C, D, E, F, G) and the set of following functional dependencies.

Marks: 2 **MSQ**

$A \rightarrow E$
$\{B, C\} \rightarrow \{G, D\}$
$D \rightarrow F$

Which of the following statement(s) is/are true?

a) $\{A, C\} \rightarrow F$

b) $\{B, D, E\} \rightarrow B$

c) $\{A, B, D\} \rightarrow C$

d) $\{A, B, C\} \rightarrow \{E, F\}$

Answer: b), d)
Explanation: a) closure of {A, C} = {A, C, E}; hence, {A, C} does not determine F.
b) is true as the functional dependency is trivial.
c) closure of {A, B, D} = {A, B, D, E, F}; hence, {A, B, D} does not determine C.
d) is true as {A, B, C} is a candidate key.

Question 5

Consider the relation: *Marks:2* **MSQ**

`Student (Stu_ID, Dept, Cls_Roll)`

The following functional dependencies are given:

```
FD1:  Stu_ID → Dept
FD2:  Dept → Stu_ID
FD3:  Stu_ID → Cls_Roll
FD4:  Cls_Roll → Stu_ID
FD5:  Dept → Cls_Roll
```

Which functional dependencies should be removed to obtain the canonical cover of the set?

a) FD2 and FD3

b) FD1 and FD5

c) FD5

d) FD1

Answer: (a), (c)
Explanation:
FD1 is not redundant as $(\texttt{Stu\_ID})^+$ without FD1 does not contain Dept. FD2 is redundant as $(\texttt{Dept})^+$ remains the same without FD2. Hence, it can be removed. Now we have FD1, FD3, FD4, FD5.
FD3 is redundant as $(\texttt{Stu\_ID})^+$ remains the same without FD3. Hence, it can be removed. Now we have FD1, FD4, FD5. The set cannot be further reduced. So, the final reduced set is FD1, FD4, FD5. If the above procedure starts by rearranging the set and placing FD5 at the beginning, the canonical cover is reduced to FD1, FD2, FD3, FD4. Hence, both options (a) and (c) are correct.

Question 5

Consider the following relation `Figure(No, Caption, Scale, Format, Color, Position)` with the following multi valued dependencies:

`FD1: No` $\rightarrow\rightarrow$ `Caption`

`FD2: Caption` $\rightarrow\rightarrow$ `{Color, Position}`

Which of the following dependencies can be derived? *Marks: 2* **MSQ**

a) `No` $\rightarrow\rightarrow$ `{Color, Position}`

b) `No` $\rightarrow\rightarrow$ `{Scale, Format, Color, Position}`

c) `{Color, Position}` $\rightarrow\rightarrow$ `No`

d) `{Scale, Format, Color, Position}` $\rightarrow\rightarrow$ `No`

Answer: a), b)

Explanation: Options a) and b) are correct according to the Complementation and Transitivity rule as described in Lecture 20, slide 13. Applying Complementation to FD1, we get option b) and applying transitivity to FD1 and FD2, we get option a). However, the reverse is not true. Hence, c) and d) are incorrect.

Question 1

Consider a relation `Transport(TID, TName, Route, MFD)` where the superkeys are as follows:

`{TID}, {TID, TName}, {Route, MFD}, {Route, MFD, TName}`.

Select the possible candidate key(s). *Marks: 2* **MSQ**

a) {TID}

b) {Route}

c) {Route, MFD}

d) {MFD}

Answer: a), c)

Explanation: Minimal superkeys are candidate keys. Here, TID alone is a superkey, hence, any superset of TID will also be a superkey. But only TID can be a candidate key. Similarly, {Route, MFD} is a superkey and so is any of its superset. But, {Route, MFD} is minimal,as, removing any one attribute from the set makes it ineligible to remain as a superkey. So {Route, MFD} is also a valid candidate key.

Hence, (a) and (c) are correct options.

Question 5

Consider the relation KEYSTORE(KID, KeyType, LockType, KeyColor, Material, Mfd) with the following functional dependencies.

FD1: {KID, KeyType}→LockType

FD2: {KeyColor, LockType}→{KID, Material}

FD3: Material→Mfd

Which of the following is (are) not the candidate key(s)? *Marks: 2* MSQ

a) {KID, KeyType, KeyColor}

b) {LockType, KeyType, KeyColor}

c) {Material, LockType, KeyColor}

d) {KeyType, KeyColor, Mfd}

Answer: c), d)

Explanation: From the given functional dependencies, it can be observed that the closure of {KID, KeyType, KeyColor} and {LockType, KeyType, KeyColor} produces all the attributes. Thus these are the candidate keys. Hence options c) and d) are correct.

Question 6

Work		
Incharge	**Department**	**Experience**
S. Roy	Production	5
A. Bera	Sales	2
S. Roy	HR	5
S. Rai	Development	3
B. Mallik	Testing	4
S. Sinha	Testing	4

Which of the following functional dependencies hold on Work? *Marks: 2* MSQ

a) Incharge → Department

b) Incharge → Experience

c) Experience → Incharge

d) Department → Experience

Answer: b), d)

Explanation: For options b), d), the attributes on the L.H.S can uniquely identify the attributes on the R.H.S.

<u>Relational Algebra</u>

Question 10

Consider the following schema: *Marks: 2* **MSQ**

R(<u>a</u>,b,c,d)
S(<u>x</u>,y,z)
T(<u>a</u>,<u>x</u>)

Four relational algebra queries are given below:

Q1: $\sigma_{R.a=101 \wedge R.c \geq 1000}((R \bowtie S) \bowtie T))$
Q2: $\sigma_{R.a=101 \wedge R.c \geq 1000}(R \bowtie (S \bowtie T))$
Q3: $\Pi_{S.x}(\sigma_{S.z=\text{'}kolkata\text{'}}(S \bowtie T))$
Q4: $\Pi_{S.x}(\sigma_{S.z=\text{'}kolkata\text{'}}(S)) \bowtie T$

Identify the correct option from the following.

a) Q1 and Q2 give different results

b) Q1 and Q2 give the same result

c) Q3 and Q4 give different results

d) Q3 and Q4 give the same result

Answer: b), c)
Explanation: Q1 and Q2 will give the same result because according to transformation rules of relational algebra natural join and Cartesian product are always associative.
Q3 and Q4 will give different results because in Q3 projection operation is performed on natural join of S and T. So, it gives the attribute x of S relation only.
But in Q4, natural join is performed between one attribute of S and all attributes of T.
So, it gives the result of one attribute of S and all attributes of T.

Question 6

Identify the correct operation which produces the below given output based on two relations R1 and R2

R1	
name	class
Raj	9
Meera	5
Sourav	12
Kirti	10

R2	
name	class
Meera	5
John	7

Output:

A	B
Meera	5

Marks: 2 **MSQ**

a) $R1 \cap R2$

b) $R1 - R2$

c) $R1 - (R1 - R2)$

d) $R1 \cup R2$

Answer: a), c)

Explanation: The common tuples between the two tables comes as output. Hence intersection between the tables has been performed. So option a) is correct. Again $R1 \cap R2 = R1 - (R1 - R2)$. Hence option c) is also correct.

Question 1

Consider the following instance:

Customer	
Name	Flower
Arnab	Daisy
Smriti	Daisy
Arushi	Orchid
Ranjit	Sunflower
Subhash	Rose
Arnab	Rose
Arnab	Orchid

Identify the correct instance(s) of `Order` such that (`Customer` ÷ `Order`) displays `Name` "Arnab" as output.

Marks: 2 **MSQ**

a)

Order
Flower
Daisy
Orchid
Rose

b)

Order
Flower
Daisy
Orchid
Sunflower

c)

Order
Flower
Daisy
Orchid

d)

Order
Flower
Daisy

Answer: a), c)
Explanation: As per the rules of division operation.

Question 5

Consider the following tables:

R_1	
Date	Total no of items sold
01/06/2020	500
02/06/2020	400
08/06/2020	300
11/06/2020	800
21/06/2020	900

R_2	
Date	Total no of items sold
21/05/2020	200
22/05/2020	400
08/06/2020	300
12/06/2020	100
21/06/2020	900

Identify the correct operation(s) which will produce the following output from the two above relations.

output table	
Date	Total no of items sold
08/06/2020	300
21/06/2020	900

Marks: 2 **MSQ**

a) $R_1 - R_2$

b) $R_1 \cap R_2$

c) $(R_1 \cup R_2) - (R_1 \cap R_2)$

d) $(R_1 \cap R_2) \cup (R_2 \cap R_1)$

Answer: b),d)
Explanation: The output corresponds to $R_1 \cap R_2$. So, option b) is correct. But option d) is equivalent to option b). Thus, option d) is also correct.

Question 7

A pet rescue organization considers the following relations: `Owner(OID, Name, Address)`, `Pet(PID, Breed)`, and `Rescue(OID, PID)`
Select the correct Relational Algebra expression(s) equivalent to the following statement.

"Display the `OID`s of those `Owners` whose `Pets` are Labradors or Beagles."

Marks: 2 **MSQ**

a) $\Pi_{\texttt{OID}}(\Pi_{\texttt{OID, PID}}(\sigma_{\texttt{Breed="Labrador"}\vee\texttt{Breed="Beagle"}}\texttt{Pet} \ltimes \texttt{Rescue}))$

b) $\Pi_{\texttt{OID}}(\sigma_{\texttt{Breed="Labrador, Beagle"}}\texttt{Pet} \ltimes \texttt{Rescue})$

c) $\Pi_{\texttt{OID}}(\sigma_{\texttt{Breed="Labrador"}\vee\texttt{Breed="Beagle"}}\texttt{Pet} \ltimes \texttt{Rescue})$

d) $\Pi_{\texttt{OID}}(\Pi_{\texttt{PID}}(\Pi_{\texttt{Breed}}\texttt{Pet}) \ltimes \texttt{Rescue})$

Answer: a), c)
Explanation: As per the syntax and semantics of Relational algebra. In option (b), $\vee$ is missing in the condition, Options (a) and (c) are the same except for left most projection operators. In consecutive projection operators, the first one from the left prevails. In option d), the given condition for `Breed` is not used.

Question 2

Consider the following relations `Zone1(Zone, Site)`, `Zone2(Zone, Site)`. The following operation is performed. `Zone2` is subtracted from `Zone1` and the difference is subtracted from `Zone1`. The following output is generated:

Zone	Site
1	Mountain
4	Sea

Identify the correct instances of `Zone1` and `Zone2`.

Marks: 2 **MSQ**

a)

Zone1	
Zone	Site
1	Mountain
4	Sea

Zone2	
Zone	Site
2	Hill
6	Sea

b)

Zone1	
Zone	Site
1	Mountain
2	Hills
3	Sea
4	Sea

Zone2	
Zone	Site
1	Mountain
4	Sea
5	Forest
6	River

c)

Zone1	
Zone	Site
1	Mountain
2	Hills
3	Sea
6	Sea

Zone2	
Zone	Site
1	Mountain
4	Sea
5	Forest
6	Sea

d)

Zone1	
Zone	Site
1	Mountain
4	Sea

Zone2	
Zone	Site
1	Mountain
4	Sea

Answer: b), d)
Explanation: The given operation is equivalent to the intersection of Zone1 and Zone2. Hence, options b) and d) are correct.

Transactions

Question 9

Suppose in a database, there are three transactions T_1, T_2 and T_3 with timestamp 10, 20 and 30 respectively. T_2 is holding a data item which T_1 and T_3 are requesting to acquire.Which of the following statement(s) is (are) correct in respect of `Wound-Wait` Deadlock Prevention scheme? *Marks: 2* **MSQ**

a) Transaction T_1 will wait for T_2 to release the data item.

b) Transaction T_2 will be aborted.

c) Transaction T_3 will wait for T_2 to release the data item.

d) Transaction T_3 will be aborted.

Answer: b), c)
Explanation: `Wound-Wait` Deadlock Prevention scheme: When `TA1` requests data item held by `TA2` (older means smaller timestamp), two cases may arise.

- If `TA1` older than `TA2` then `TA1` wounds `TA2` (`TA2 will be aborted`).
- If `TA1` younger than `TA2` then `TA1` will wait to release the data item held by TA2.

In this case `Transaction` T_1 is older and T_3 is younger in respect of `Transaction` T_2. Hence, options b) and c) are correct.

Question 10

Identify the correct statement(s) about the `lock compatibility matrix` given below. *Marks:2* **MSQ**

	S	X
S	True	False
X	False	False

a) Data item can be both read and written using `X-lock`.

b) If a transaction holds an `S-lock` on an item, other transactions will not be allowed to obtain a `S-lock` on the same item.

c) Data item can be both read and written using `S-lock`.

d) If a transaction holds an `X-lock` on an item, other transactions can not allowed to obtain a `X-lock` on the same item.

Answer: a), d)
Explanation: As per lock based protocols (Refer Module 34 slide 10).

Question 1

Which of these is/are **false** about transaction?

Marks: 2 **MSQ**

a) Multiple transactions can execute concurrently whether they are isolated or not.

b) If a transaction completes successfully, anything it changes in the database must persist even if system failure occurs.

c) Either all operations of a transaction are executed properly or none.

d) After completion of the transaction database might be consistent or inconsistent.

Answer a), d)

Explanation: According to `ACID` properties of transaction, two transaction can execute concurrently only if they are isolated. Hence option a) is false. After completion of transaction, database must be consistent. Hence option d) is false.

Question 10

Consider the **student** table.

Marks: 2 **MSQ**

student		
regd_no	name	dept
1	SAM	CSE
5	ANIEE	ECE
1	ALEX	CIVIL
10	JOHN	CSE

Find which of the following statement/s is/are correct for the records selected below from the **student** table.

regd_no	name	dept
1	SAM	CSE
10	JOHN	CSE

a) Find the students who belong to 'CSE' department.

b) Find the names of the students who belong to 'CSE' department.

c) Find the students whose name is either 'SAM' or 'JOHN'.

d) Find the students whose registration number is either 1 or 10.

Answer: (a), (c)

Explanation: Options (b) and (d) are incorrect, since option (b) finds only the name of students who belongs to 'CSE' department and not the other attributes and option (d) finds the students whose registration number is either 1 or 10 which also includes the tuple `(1, ALEX, CIVIL)`. Hence, (a) and (c).

Question 2

Identify the correct `INSERT` command/s to create a record for the `customer` table as given below. Primary key is underlined in the schema.

Marks: 2 **MSQ**

customer		
customer_name	customer_street	customer_city
P. SAMBHU	NEW COMPLEX	CHENNAI

a)
```
INSERT INTO customer
VALUES('P. SAMBHU', 'NEW COMPLEX', 'CHENNAI');
```

b)
```
INSERT INTO customer
VALUES("P. SAMBHU", "NEW COMPLEX", "CHENNAI");
```

c)
```
INSERT INTO customer
VALUE('P. SAMBHU', 'NEW COMPLEX', 'CHENNAI');
```

d)
```
INSERT INTO customer
(customer_name,customer_street,customer_city)
 VALUES('P. SAMBHU', 'NEW COMPLEX', 'CHENNAI');
```

Answer: a), d)

Explanation: The syntax of INSERT command is:

Question 4

Which of the following option/s is/are correct to find the names of the customers who are having account in the bank without having a loan in the bank? Primary keys are underlined in the schemas.

Marks: 2 **MSQ**

- depositor(<u>customer_name</u>, <u>account_number</u>) – has account
- borrower(<u>customer_name</u>, <u>loan_number</u>) – has taken loan

a)
```
SELECT DISTINCT customer_name
FROM depositor INTERSECT
SELECT DISTINCT customer_name
FROM borrower;
```

b)
```
SELECT customer_name
FROM depositor
WHERE customer_name
NOT IN
(SELECT customer_name
FROM borrower);
```

c)
```
SELECT customer_name
FROM borrower
WHERE customer_name
NOT IN
(SELECT customer_name
FROM depositor);
```

d)
```
SELECT DISTINCT customer_name
FROM depositor
MINUS
SELECT DISTINCT customer_name
FROM borrower;
```

Answer: b), d)

Explanation: We can use `NOT IN` or `MINUS` or `EXCEPT` operators to solve this type of query.

So option b) and d) are correct and rests are incorrect.

Option a) finds the names of the customers who have both an account and a loan in the bank.

Option c) finds the names of the customers who have a loan but do not have an account in the bank.

Question 5

Consider the schemas **person** and **salary** to identify the correct query/queries that display/s the name and the salary of the persons who work in 'KOLKATA'. Primary keys are underlined in the schemas. *Marks: 2* **MSQ**

- `person(person_id, person_name, location)`
- `salary(person_id, sal)`

a)
```
SELECT person_name, sal
FROM person,salary
WHERE person.person_id = person_id
AND location = 'KOLKATA';
```

b)
```
SELECT person_name, sal
FROM person,salary
WHERE location = 'KOLKATA';
```

c)
```
SELECT person_name, sal
FROM person,salary
WHERE person.person_id = salary.person_id
AND location = 'KOLKATA';
```

d)
```
SELECT person_name, sal
FROM person JOIN salary
ON person.person_id = salary.person_id
AND location = 'KOLKATA';
```

Answer: c), d)

Explanation: From the above option c) and d) are correct as it uses join operation to compare the attributes `person.person_id` and `salary.person_id` to avoid ambiguity in the columns.

Question 6

Which is/are the correct SQL command/s to display the `loan_numbers` of the customers in ascending order of their `loan_amounts`. Primary key is underlined in the schemas.

Marks: 2 **MSQ**

- loan(<u>loan_number</u>, branch_name, amount)

a) `SELECT loan_number FROM loan ORDER BY amount;`

b) `SELECT loan_number FROM loan SORT BY amount;`

c) `SELECT loan_number FROM loan ORDER BY amount ASC;`

d) `SELECT loan_number FROM loan GROUP BY amount;`

Answer: a), c)

Explanation: The `ORDER BY` clause is used to sort list items in ascending order by default. We also use DSC or ASC to sort the values in descending or ascending order respectively.

Option b) and d) are incorrect since option b) generates the error message, SQL command not properly ended since SORT BY is not valid for this.

Option d) also generates error message, not a GROUP BY expression.

So option a) and c) are correct.

Question 9

Identify the following the correct way/s to delete all rows at a time from `salary` table.

Marks: 2 **MSQ**

a) `DELETE * FROM salary;`

b) `TRUNCATE TABLE salary;`

c) `DELETE FROM salary;`

d) `TRUNCATE TABLE FROM salary;`

Answer: b), c)

Explanation: To delete all rows from a table we can use either `DELETE` or `TRUNCATE` command and the syntax for these commands are:

```
DELETE FROM table name
[WHERE conditions];
```

Here condition is optional.

and

```
TRUNCATE TABLE table name;
```

So option b) and c) are correct and rest of the options are incorrect.

Option a) gives an error due to * operator in `DELETE` command.

Option d) also gives an error as it violates the syntax of `TRUNCATE` command.

Indexing

Question 1

Secondary indices are:

Marks: 2 MSQ

a) Clustering index

b) Non-Clustering index

c) Sparse index

d) Dense index

Answer: b), d)

Explanation: Secondary indices are also called Non clustering index and Secondary index should be Dense index.